$$1 + 3 + (-1.4) + 7 + 4 + 1 + 2 + \underline{1 + 3 + (-1.4) + 7 + 4}$$

You can see the repeated elements underlined in the formula above. So, for $n=7$, we need 11 operations.

In general, we will need $n+4$ operations, and $T_1(n) = n+4$.

Is it true that $T_1(n) = O(n)$? Let's go back to our definition: we need to find two constants, an integer n_0 and a real number c, such that:

$T_1(n) \leq c * n$ for each $n > n_0$

$\Leftrightarrow$

$n + 4 \leq c * n$ for each $n > n_0$

$\Leftrightarrow$

$c \geq 1 + 4/n$ for each $n > n_0$

Since $4/n$ decreases when n grows, we can choose $c=2$ and $n_0=4$, or (say) $c=100$ and $n_0=1$, and the inequality will be satisfied.

Same if we consider summing squares: for n numbers, we will need n multiplications (or squarings) and $n-1$ sums, so $T_2(n) = 2n-1$.

We can prove that $T_2(n) = O(n)$ by choosing $c=2.5$ and $n_0=1$.

Now, we could also say that $T_1(n) = O(n^2)$: we leave you as an exercise to find appropriate values for c and n_0. However, this bound would not be strict: in other words, there exist